Quantitative Confidentiality for Cipher Maps

Alexander Towell
lex@metafunctor.com

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Abstract

A cipher map is a total function on bit strings implementing a trapdoor approximation of a latent function: the untrusted machine evaluates it blindly; only the trusted machine, holding the secret, can encode inputs and decode outputs. We develop a quantitative confidentiality theory for cipher map systems. The central measure is the *entropy ratio* $e = H/H^*$, where H is the observed entropy of cipher map outputs and H^* is the maximum entropy achievable under system constraints. This ratio lies in $[0, 1]$: zero means fully predictable; one means indistinguishable from uniform. We identify three levers for improving confidentiality—noise injection, multiple representations, and encoding granularity—and prove how each affects the entropy ratio through the cipher map parameters $(\eta, \varepsilon, \delta, \mu)$. We analyze confidentiality under composition, showing how shared variables and Boolean chain operations create correlation leakage, and provide practical measurement techniques including compression-based entropy estimation. Experimental validation on encrypted Boolean search over the 20 Newsgroups corpus confirms the theoretical predictions for FPR compounding and the encoding granularity spectrum.

1 Introduction

Given a cipher map system—a trusted machine encoding queries, an untrusted machine evaluating total functions on bit strings, and results decoded only by the trusted machine—how much does the untrusted machine learn?

The cipher map framework [4] provides four properties (totality, representation uniformity, correctness, composability) parameterized by $(\eta, \varepsilon, \delta, \mu)$ that characterize what the untrusted machine can and cannot do. Totality ensures every input produces output; representation uniformity bounds frequency analysis; correctness governs error rates; composability enables chained evaluation. But these properties are *qualitative* guarantees. A system designer needs *quantitative* answers: given a specific deployment with a known query distribution, known vocabulary size, and known resource budget, how confidential is the system, and how can it be improved?

This paper develops the quantitative theory. The central idea is the *entropy ratio*: the ratio of observed entropy in the cipher map output stream to the maximum entropy achievable under system constraints. We ground this idea in the cipher map framework, connecting the entropy ratio to the four cipher map parameters, and identify three levers for improving confidentiality:

1. **Noise injection.** The trusted machine mixes R filler queries per N real queries. By totality, the untrusted machine cannot distinguish real from filler. This dilutes the adversary's correlation estimates.

2. **Multiple representations.** Each element receives $K(x) \geq 1$ encodings, with $K(x) \propto 1/D(x)$ to flatten the cipher value distribution. This controls the representation uniformity parameter δ.

3. **Encoding granularity.** The entanglement parameter p controls how many correlated values are encoded as a single unit. Joint encoding hides correlations; component-wise encoding leaks them. The spectrum runs from full correlation hiding ($p = k$, space $O(|Y|^k)$) to no correlation hiding ($p = 1$, space $O(k \cdot |Y|)$).

We then analyze how confidentiality behaves under composition: when cipher maps are chained (Boolean AND/OR/NOT over cipher sets), FPR compounds predictably but correlations between shared variables create leakage that marginal uniformity alone does not prevent.

Finally, we provide practical measurement tools. The entropy of a cipher map output stream can be estimated by compressing it: more compressible means more predictable means less confidential. This avoids the need for explicit probabilistic models of the adversary.

Contributions.

1. The entropy ratio $e = H/H^*$ as a quantitative confidentiality measure for cipher map systems, with a precise relationship to the cipher map parameters (§3).
2. Theorems connecting each of the three levers (noise injection, multiplicity, granularity) to the entropy ratio, with explicit cost/benefit trade-offs (§4).
3. Analysis of confidentiality under composition: FPR compounding through Boolean chains and correlation leakage from shared variables (§5).
4. Practical measurement via compression-based entropy estimation and a case study demonstrating confidentiality improvement from 59% to 85% (§6, §7).

What this paper does not do. We do not use ORAM, FHE, or simulation-based security definitions. Privacy in the cipher map model comes from the one-way hash, the totality of $\hat{f}$, and representation uniformity—not from access-pattern indistinguishability or computational hardness assumptions beyond the hash function. The framework is information-theoretic throughout.

2 Preliminaries

We recall the cipher map abstraction from [4]. All notation follows that paper; we cite rather than re-derive.

Definition 2.1 (Cipher map [4, Def. 3.1])**.** A *cipher map* for a latent function $f : X \to Y$ is a tuple $(\hat{f}, \mathrm{enc}, \mathrm{dec}, s)$ where $\hat{f} : \{0,1\}^n \to \{0,1\}^n$ is a total function on n-bit strings, $\mathrm{enc} : X \times \{0, \ldots, K(x)-1\} \to \{0,1\}^n$ is an encoding function with $K(x) \geq 1$ representations per element, $\mathrm{dec} : \{0,1\}^n \to Y \cup \{\perp\}$ is a decoding function, and s is a secret (the trapdoor). The untrusted machine holds $\hat{f}$. The trusted machine holds enc, dec, and s.

The four properties, parameterized by $(\eta, \varepsilon, \mu, \delta)$ [4, Sec. 4]:

Totality. $\hat{f}$ is defined on all 2^n inputs. Out-of-domain outputs are indistinguishable from uniform under the random oracle model.

Representation uniformity (δ-bounded). The marginal distribution of cipher values is δ-close to uniform over $\{0,1\}^n$ in total variation distance: $d_{\mathrm{TV}}(P, U) = \frac{1}{2}\sum_c |P(c) - U(c)| \leq \delta$. Informally, δ bounds the advantage of any test that tries to distinguish real cipher values from random ones.

Correctness (η-bounded). The fraction of domain elements that decode incorrectly is at most η:

$$\Pr[\mathrm{dec}(\hat{f}(\mathrm{enc}(x,k))) \neq f(x)] \leq \eta.$$

Nonzero η provides plausible deniability (an observed result might be wrong) and reduces construction cost.

Noise-decode probability (ε). A random n-bit string decodes to a valid output with probability ε. Lower ε means out-of-domain queries are more likely to be rejected (decoded as $\perp$), at the cost of more bits per element.

Value cost ($\mu = H(Y)$). The average number of bits needed to encode one output value, equal to the Shannon entropy of the output distribution.

Composability. The composition of two cipher maps is again a cipher map: $\hat{g} \circ \hat{f}$ has correctness $\eta_{g \circ f} \leq 1 - (1-\eta_f)(1-\eta_g)$. Errors accumulate predictably, enabling pipeline analysis.

The space per element decomposes as bits/element $= -\log_2 \varepsilon + H(Y)$, where $-\log_2 \varepsilon$ bits distinguish signal from noise and $H(Y) = \mu$ bits encode the function value [4, Thm. 6.1].

Trusted/untrusted model. The trusted machine T encodes inputs, decodes results, and injects filler queries. The untrusted machine U evaluates $\hat{f}$ on any bit string and returns results. U cannot decode, distinguish real from filler queries, determine the domain X, or enumerate which inputs are "in-domain" [4, Sec. 5].

Acceptance predicates. The batch construction of a cipher map is unified under an *acceptance predicate*: a family of disjoint sets $\{A(y) \subseteq \{0,1\}^n : y \in Y\}$ that partitions hash space among output values [4, Sec. 6.2]. Shannon-optimal allocation sets $|A(y)| \propto \Pr[f(X) = y]$, simultaneously minimizing space and maximizing output indistinguishability.

Information-theoretic notation. We use standard information-theoretic quantities throughout. *Shannon entropy* $H(X) = -\sum_x p(x) \log_2 p(x)$ measures the average surprise (in bits) of a random variable; higher entropy means less predictable. *Conditional entropy* $H(X \mid Y)$ is the remaining uncertainty in X after observing Y. *Mutual information* $I(X;Y) = H(X) - H(X \mid Y)$ quantifies how much observing Y reveals about X; zero means independence. *KL divergence* $D_{\mathrm{KL}}(P \| Q) = \sum_x P(x) \log_2(P(x)/Q(x))$ measures how far P is from Q; it is zero iff $P = Q$ and connects to the entropy ratio via $H(P) = H^* - D_{\mathrm{KL}}(P \| U) \cdot H^*$ when U is the maximum-entropy distribution. We use D for the distribution of queries over the domain X, and Q for the induced distribution over cipher values. All logarithms are base 2.

3 The Confidentiality Measure

3.1 Observed and Maximum Entropy

Consider a cipher map system in which the trusted machine submits a stream of cipher values to the untrusted machine. The untrusted machine observes a sequence $c_1, c_2, \ldots, c_N$ of n-bit strings, each either a real encoding $\mathrm{enc}(x_i, k_i)$ or a filler string drawn uniformly from $\{0,1\}^n$.

Definition 3.1 (Observed entropy)**.** The *observed entropy* H of a cipher map system is the Shannon entropy of the joint distribution over the observable sequence:

$$H = H(C_1, C_2, \ldots, C_N), \tag{1}$$

where each C_i is the random variable corresponding to the i-th cipher value observed by the untrusted machine.

Definition 3.2 (Maximum entropy under constraints)**.** The *maximum entropy* H^* is the maximum of H over all distributions on the observable sequence that are consistent with the system constraints: domain size $|X|$, vocabulary size $|Y|$, query arrival rate λ, mean query size μ_q, number of agents k, and the cipher map parameters $(\eta, \varepsilon, \delta)$.

By the maximum entropy principle [1], H^* is achieved when the observable distributions are as uniform as possible subject to the constraints. For a single cipher value drawn from a vocabulary of size m, the maximum entropy is $\log_2 m$ (uniform distribution). For a sequence of N independent cipher values, the maximum entropy is $N \log_2 m$.

3.2 The Entropy Ratio

Definition 3.3 (Entropy ratio)**.** The *entropy ratio* of a cipher map system is

$$e = \frac{H}{H^*} \in [0, 1]. \tag{2}$$

The entropy ratio has a direct operational interpretation:

- $e = 0$: the output sequence is fully predictable. The untrusted machine can predict every cipher value with certainty. No confidentiality.
- $e = 1$: the output sequence is indistinguishable from the maximum entropy distribution under system constraints. The untrusted machine's best prediction is no better than guessing from the maximum entropy distribution. Maximum confidentiality.

3.3 Relationship to Cipher Map Parameters

The entropy ratio connects to the four cipher map parameters through the following result.

Theorem 3.1 (Entropy ratio decomposition)**.** *Let $(\hat{f}, \mathrm{enc}, \mathrm{dec}, s)$ be a cipher map for $f : X \to Y$ with parameters $(\eta, \varepsilon, \delta, \mu)$. Let D be the query distribution on X, and let Q be the induced distribution on cipher values. Then the per-query observed entropy satisfies:*

$$H(Q) = H^* - D_{\mathrm{KL}}(Q \| U) \cdot H^* = H^*(1 - \delta'), \tag{3}$$

where U is the uniform distribution on $\{0,1\}^n$ and δ' captures the deviation from uniformity. In particular:

1. *If $\delta = 0$ (perfect representation uniformity), then $Q = U$ and $H(Q) = H^* = n$ bits, giving $e = 1$.*
2. *If $K(x) = 1$ for all x (simple substitution, no multiple representations), then Q inherits the non-uniformity of D, and $e \leq H(D)/\log_2 |X|$.*

3. *The entropy ratio is bounded by* $e \geq 1 - \delta \cdot \log_2(2^n/\delta)$ *for small* δ*, via Pinsker's inequality.*

Proof. For part (1): when $\delta = 0$, Q is uniform over $\{0,1\}^n$ by definition of representation uniformity. The entropy of the uniform distribution on 2^n elements is n bits, which equals H^*.

For part (2): with $K(x) = 1$, the encoding enc is injective, so Q is a relabeling of D. Entropy is invariant under bijection, giving $H(Q) = H(D)$. The maximum over distributions on $|X|$ elements is $\log_2 |X|$, so $e = H(D)/\log_2 |X|$.

For part (3): by Pinsker's inequality (which bounds KL divergence from below by squared TV distance), $d_{\mathrm{TV}}(Q, U) \leq \sqrt{D_{\mathrm{KL}}(Q\|U)/2}$. Since $d_{\mathrm{TV}}(Q, U) \leq \delta$, we have $D_{\mathrm{KL}}(Q\|U) \leq 2\delta^2$. Then $H(Q) = \log_2 2^n - D_{\mathrm{KL}}(Q\|U) \geq n - 2\delta^2$, and $e \geq 1 - 2\delta^2/n$. □

Remark 3.1 (The role of ε). The noise decode probability ε does not appear directly in the per-query entropy, but it determines the space cost: achieving small ε requires $-\log_2 \varepsilon$ bits per element. The indirect effect is that larger ε means a denser cipher space (more random bit strings decode to valid outputs), which helps uniformity but increases false positives.

Remark 3.2 (The role of η). The correctness parameter η provides a form of plausible deniability. With $\eta > 0$, the untrusted machine cannot be certain that a decoded output reflects the true function value. This adds $H_b(\eta) = -\eta \log_2 \eta - (1-\eta)\log_2(1-\eta)$ bits of uncertainty per query (the binary entropy of the error event).

4 Three Levers for Improving Confidentiality

4.1 Noise Injection

The trusted machine can mix R filler queries with N real queries. By totality, the untrusted machine cannot distinguish real from filler: both are n-bit strings, and $\hat{f}$ produces n-bit output on both.

Theorem 4.1 (Noise dilution)**.** *Let the trusted machine submit* N *real queries drawn from distribution* D *and* R *filler queries drawn uniformly from* $\{0,1\}^n$*. The mixed sequence has* $N + R$ *elements, each of which is real with probability* $\rho = N/(N+R)$ *and filler with probability* $1 - \rho$*. Then:*

1. *The per-element entropy of the mixed distribution is*

$$H_{\mathrm{mix}} = H_b(\rho) + \rho \cdot H(D) + (1-\rho) \cdot n, \tag{4}$$

   where $H_b(\rho)$ *is the binary entropy of* ρ*.*

2. *As* $R/N \to \infty$*,* $H_{\mathrm{mix}} \to n = H^*$ *and* $e \to 1$*.*

3. *The adversary's estimate of any correlation between query elements has variance* $O(1/(N+R))$*. Mixing with* $R \gg N$ *filler dilutes the real signal proportionally: the adversary needs* $\Omega((1+R/N)^2)$ *observations to achieve the same estimation accuracy as observing* N *pure real queries.*

Proof. For part (1): each observed element is a mixture. With probability ρ it is drawn from D (entropy $H(D)$); with probability $1-\rho$ it is uniform on $\{0,1\}^n$ (entropy n). By the chain rule, the per-element entropy is the entropy of the mixing indicator plus the conditional entropy given the indicator.

For part (2): as $\rho \to 0$, the distribution converges to uniform on $\{0,1\}^n$.

For part (3): the adversary observes a mixture. For any statistic T computed from the observed sequence, the contribution of real queries is diluted by the noise fraction. Specifically, let $\hat{\theta}$ be the adversary's estimator of a parameter θ of D. Each observation contributes information about θ only when it is a real query (probability ρ). The effective sample size for estimating θ is $\rho \cdot (N+R) = N$, but the adversary does not know which observations are real. The best the adversary can do is treat each observation as drawn from the mixture, giving Fisher information reduced by a factor of ρ^2 relative to observing D directly. □

Cost. Noise injection costs bandwidth: the total query volume increases by a factor of $(N + R)/N = 1 + R/N$. It does not cost space (no additional cipher map storage). A noise ratio of $R/N = 1$ (equal filler and real queries) doubles bandwidth; $R/N = 9$ (90% filler) increases bandwidth tenfold but brings e close to 1 for most practical distributions.

4.2 Multiple Representations ($K > 1$)

Each element x can be given $K(x) \geq 1$ distinct encodings. The trusted machine, when encoding x, chooses k uniformly from $\{0, \ldots, K(x) - 1\}$. Assigning $K(x) \propto 1/D(x)$ equalizes the cipher value frequencies, achieving $\delta \approx 0$.

Theorem 4.2 (Representation uniformity via multiplicity)**.** *Let D be a distribution on X with $D(x) > 0$ for all x. Set $K(x) = \lceil c/D(x) \rceil$ for a normalizing constant c. Then the induced distribution on cipher values satisfies:*

$$d_{\mathrm{TV}}(Q, U) \leq \frac{|X|}{2 \sum_x K(x)}. \tag{5}$$

In particular, $d_{\mathrm{TV}}(Q, U) \to 0$ as $c \to \infty$.

Proof. The induced distribution on cipher values is $Q(v) = \sum_x D(x) \cdot |\{k : \mathrm{enc}(x, k) = v\}|/K(x)$. Under the random oracle model, each encoding maps to a distinct cipher value (with high probability when n is large), so the sum simplifies: each cipher value v is hit by exactly one pair (x, k), giving $Q(v) = D(x)/K(x)$ for the (x, k) mapping to v. With $K(x) = \lceil c/D(x) \rceil$, we have $D(x)/K(x) \approx c$ for all x. The total number of cipher values is $\sum_x K(x)$, and the uniform distribution on that many values assigns probability $1/\sum_x K(x)$ to each. The TV distance between Q and uniform is bounded by the maximum deviation, which is at most $|X|/(2\sum_x K(x))$ by the rounding error in $\lceil c/D(x) \rceil$. □

Cost. Multiple representations cost space: each element now occupies $K(x)$ slots in the cipher map, so the total space is $\sum_x K(x) \cdot (-\log_2 \varepsilon + H(Y))$ bits. For a Zipf distribution $D(x) \propto 1/i$ on m elements, the total number of representations is $\sum_x K(x) \approx c \sum_x 1/D(x) = c \cdot m \cdot H_m$ where $H_m = \sum_{i=1}^{m} 1/i$ is the m-th harmonic number.

Example 4.1 (Homophonic encryption for Zipf-distributed queries)*.* A vocabulary of $m = 10{,}000$ words with Zipf distribution $D(x_i) \propto 1/i$ has entropy $H(D) \approx 7.83$ bits, compared to $H^* = \log_2 10{,}000 \approx 13.29$ bits, giving $e \approx 0.59$. Flattening the top $b = 100$ words via $K(x_i) = \lceil D(x_1)/D(x_i) \rceil$ (approximately i representations for the i-th ranked word) increases the effective entropy to ≈ 10.2 bits, giving $e \approx 0.77$. The space overhead is approximately 518 additional trapdoors per document.

4.3 Encoding Granularity

Representation uniformity is a marginal property: it says the cipher value distribution for each individual element is close to uniform. It says nothing about the joint distribution of multiple cipher values. The *encoding granularity* controls what correlations are hidden.

Definition 4.1 (Entanglement parameter [4, Sec. 9])**.** For a system encoding k correlated values, the *entanglement parameter* p is the number of values encoded as a single cipher map unit. The system has type $\text{cipher}(\{0,1\}^p)^{k/p}$, where each block of p values is encoded jointly.

Proposition 4.3 (Granularity spectrum [4, Prop. 9.1])**.** *For a cipher map encoding a pair of values $(a, b) \in A \times B$:*

1. ***Joint encoding*** *($p = 2$): the pair (a, b) is encoded as a single cipher value. The untrusted machine cannot distinguish (a_1, b_1) from (a_2, b_2) (up to δ), but projections require cipher map constructions by the trusted machine.*

2. ***Component-wise encoding*** *($p = 1$): a and b are encoded independently. Projections are free, but the joint distribution of (a, b) is preserved in the cipher pair. With N observations, the adversary estimates the joint distribution to accuracy ξ in TV distance from $O(|A|\cdot|B|/\xi^2)$ samples.*

The encoding granularity creates a spectrum of confidentiality/ functionality trade-offs, analyzed in detail in [3]:

Granularity	Intermediates	Correlation hiding	Space	Projections
Root ($p = k$)	0	Full	$O(\lvert Y\rvert^k)$	None free
Intermediate	few	Partial	moderate	Some free
Leaf ($p = 1$)	k	None (marginal only)	$O(k \cdot \lvert Y\rvert)$	All free

The sum-type impossibility theorem [3, Thm. 4.1] shows that this trade-off is fundamental for sum types: for $A+B$, tag hiding and untrusted pattern matching are mutually exclusive. Products leak correlations; sums leak the tag.

Confidentiality bound from orbit closure. The orbit closure [3, Sec. 5] quantifies the dynamic confidentiality cost of exposing operations. Given cipher maps $F = \{\hat{f}_1, \ldots, \hat{f}_m\}$ available to the untrusted machine and a starting cipher value c, the *orbit* $\text{orbit}_F(c)$ is the set of all cipher values reachable from c by composing operations in F. The *confidentiality* $\text{conf}_F(c)$ measures how much uncertainty the adversary retains about the latent value encoded by c: it is 1 when the adversary has learned nothing and 0 when the latent value is uniquely determined. The orbit size bounds it [3, Thm. 5.3]:

$$\text{conf}_F(c) \geq 1 - \frac{|\,\text{orbit}_F(c)|}{2^n}. \tag{6}$$

A larger orbit (more reachable cipher values) means the adversary has explored more of the cipher space, leaving fewer possibilities for the latent value. Fewer exposed operations means smaller orbit means higher confidentiality. This provides a formal justification for the root encoding strategy: exposing no intermediate cipher values gives $|\,\text{orbit}_\emptyset(c)| = 1$ and confidentiality $\geq 1 - 2^{-n}$.

5 Compositional Confidentiality

When cipher maps compose—$\hat{g} \circ \hat{f}$, or Boolean combinations like $\widehat{\text{AND}}(\hat{f}(c), \hat{g}(c))$—confidentiality behaves differently from the single-query case. Two effects interact: error compounding (correctness degrades) and correlation leakage (intermediate values reveal structure).

5.1 Error Compounding in Boolean Chains

For Boolean-valued cipher maps (set membership predicates), the composition behavior depends on the gate type. Let p_T denote the *false positive rate* (FPR):

$$p_T = \Pr[\text{dec}(\hat{f}(\text{enc}(x, k))) = \text{True} \mid f(x) = \text{False}],$$

the probability that a non-member is incorrectly decoded as True. This equals the fraction of hash space allocated to the True region in the cipher Boolean type.

Theorem 5.1 (FPR compounding [4, Sec. 8])**.** *For k independent cipher set membership tests composed via Boolean operations:*

$$\textit{AND of } k \textit{ tests:} \quad \text{FPR}_{\text{total}} = p_T^k, \tag{7}$$

$$\textit{OR of } k \textit{ tests:} \quad \text{FPR}_{\text{total}} = 1 - (1 - p_T)^k. \tag{8}$$

AND drives false positive rate down exponentially: each additional conjunct is an independent filter. OR drives it up: each additional disjunct adds false positive mass. NOT is approximate due to complement non-preservation via the pigeonhole principle [4].

Effect on confidentiality. AND chains improve precision (fewer false positives) but maintain the same recall (true positives pass all tests). From a confidentiality perspective, AND chains reduce the effective noise level: with $p_T = 0.05$ and $k = 3$, the FPR drops to $0.05^3 \approx 0.000125$, meaning almost all decoded "true" outputs correspond to genuine members. This makes the system more predictable to the adversary (lower e) in exchange for higher precision.

OR chains increase the noise level but reduce recall. This makes the system less predictable (higher e) but also less useful (more false positives in the results).

5.2 Correlation Leakage from Shared Variables

Even when each cipher map has uniform marginal output, evaluating multiple cipher maps on the same cipher value leaks correlations [4, Sec. 9.2]. If the untrusted machine evaluates $\hat{f}_1(c)$ and $\hat{f}_2(c)$ for the same c, the pair $(\hat{f}_1(c), \hat{f}_2(c))$ is a deterministic function of c, and the joint distribution preserves the latent correlation between f_1 and f_2.

Proposition 5.2 (Compositional leakage)**.** *Let $\hat{f}_1, \hat{f}_2$ be cipher maps for latent $f_1 : X \to Y_1$, $f_2 : X \to Y_2$, each with Shannon-optimal acceptance predicates. If the untrusted machine observes both $\hat{f}_1(c)$ and $\hat{f}_2(c)$ for the same cipher value c, then:*

1. *Each marginal output is individually uniform (by Shannon-optimal allocation).*
2. *The joint distribution $(\hat{f}_1(c), \hat{f}_2(c))$ preserves all correlations between $f_1(x)$ and $f_2(x)$.*
3. *A third cipher map $\hat{f}_3$ taking this correlated pair as input inherits the non-uniformity, even if $\hat{f}_3$'s acceptance predicate is Shannon-optimal for the marginal distribution of its output.*

Mitigation strategies. Two approaches, both costly [4, Sec. 9.2]:

Build the composition directly. Construct a single cipher map for $g(x) = f_3(f_1(x), f_2(x))$. The untrusted machine sees only input and final output; no intermediates are exposed. Cost: the domain is the full Cartesian product, so space is $O(|X| \cdot (-\log_2 \varepsilon + H(W)))$.

Compose components and inject noise. Keep practical component-wise evaluation and add R noise queries per N real queries. Cost: bandwidth, not space. The adversary's correlation estimates are diluted by the noise fraction.

6 Practical Measurement

6.1 Compression-Based Entropy Estimation

The entropy of a sequence can be estimated without an explicit probabilistic model by compressing it. This follows from Shannon's source coding theorem [2]: the expected length of the output of an optimal lossless compressor equals the entropy of the source.

Proposition 6.1 (Compression estimator)**.** *Let* $\mathbf{c} = (c_1, \ldots, c_N)$ *be a sequence of cipher values observed by the untrusted machine, encoded as a bit string via some fixed encoding* Encode*. Let* Compress *be a lossless compressor. Then:*

$$\hat{H} = \frac{|\mathrm{Compress}(\mathrm{Encode}(\mathbf{c}))|}{N} \tag{9}$$

is a positively biased estimator of the per-element entropy $H(C_1, \ldots, C_N)/N$*. That is,* $\hat{H} \geq H/N$ *in expectation, with equality when* Compress *is optimal.*

Proof. An optimal compressor produces output whose expected bit length equals $H(C_1, \ldots, C_N)$ by Shannon's source coding theorem. Any sub-optimal compressor produces output at least as long. The particular encoding is irrelevant: the compressor compensates for the encoding overhead (as long as the encoding is uniquely decodable). □

The compression estimator has three practical advantages:

1. It requires no model of the query distribution.
2. It captures all forms of structure (frequency, correlation, temporal patterns) that a lossless compressor can detect.
3. It is cheap to compute: a single pass through the data with `gzip` or `zstd`.

Entropy ratio estimator. Given the compression-based estimate $\hat{H}$ and a computable H^* (from the system parameters), the entropy ratio is estimated as:

$$\hat{e} = \frac{\hat{H}}{H^*}. \tag{10}$$

Since $\hat{H}$ is positively biased, $\hat{e}$ is also positively biased (overestimates confidentiality). This is the conservative direction: the system is at least as confidential as the estimate suggests.

6.2 Monte Carlo Estimation

When the query distribution is available (e.g., from historical logs), entropy can be estimated by sampling. Generate M independent query sequences of length N, compute the cipher value distribution, and estimate H directly:

$$\hat{H}_{\mathrm{MC}} = -\sum_{v} \hat{p}(v) \log_2 \hat{p}(v), \tag{11}$$

where $\hat{p}(v)$ is the empirical frequency of cipher value v across the M samples. This estimator is negatively biased for finite M (the Miller–Madow correction adds $(\hat{m} - 1)/(2M \ln 2)$ where $\hat{m}$ is the number of distinct observed values), but converges to the true entropy as $M \to \infty$.

6.3 The Leakage Analyzer

We define the *leakage analyzer* as the procedure that combines compression-based and Monte Carlo estimation:

1. Observe or simulate a sequence of cipher values.
2. Compute $\hat{H}$ via compression and $\hat{H}_{\mathrm{MC}}$ via sampling (if the distribution is available).
3. Compute H^* from the system parameters using the maximum entropy formulas.
4. Report $\hat{e} = \hat{H}/H^*$ and the entropy gap $H^* - \hat{H}$ in bits.

The entropy gap quantifies how many bits of information per query the adversary potentially gains from the non-uniformity of the system.

7 Experimental Results

We validate the theoretical predictions using the `cipher-maps` Python library, which implements the batch cipher map construction with perfect hash functions. All experiments use the 20 Newsgroups corpus (18,266 documents, 58,903 unique words).

7.1 Boolean Search: Precision and Recall

Cipher sets are built for each document using 8-bit cipher Booleans ($p_T = 0.05$, $p_F = 0.90$, $p_N = 0.05$). Table 1 shows precision and recall for Boolean queries at 5,000 documents.

Table 1: Boolean search results on 20 Newsgroups (5,000 documents).

Query	Precision	Recall	FP	FN
Single term	0.39	1.00	248	0
2-term AND	0.76	1.00	12	0
3-term AND	0.97	1.00	1	0
2-term OR	0.25	0.97	310	8
OR-AND-NOT	0.31	0.88	285	15

The results confirm the FPR compounding theory: AND drives false positives down ($248 \to 12 \to 1$), while OR and NOT introduce noise. From a confidentiality perspective, the single-term query has the highest entropy ratio (most noise in the results), while the 3-term AND has the lowest (almost deterministic results).

7.2 FPR Compounding

Table 2 compares empirical and theoretical FPR for AND and OR chains of length $k = 1$ through 5, each with base FPR $p_T = 0.05$.

Table 2: FPR compounding through Boolean chains ($p_T = 0.05$).

	AND chain			OR chain		
k	Empirical	Theory	Ratio	Empirical	Theory	Ratio
1	0.050	0.050	1.00	0.050	0.050	1.00
2	0.003	0.0025	1.20	0.095	0.098	0.97
3	0.000	0.000125	—	0.142	0.143	0.99
4	0.000	0.0000063	—	0.186	0.185	1.01
5	0.000	0.0000003	—	0.224	0.226	0.99

The AND chain FPR drops below measurable levels by $k = 3$ (as predicted by $0.05^3 \approx 10^{-4}$). The OR chain FPR matches $1 - (1 - 0.05)^k$ closely across all chain lengths.

7.3 Encoding Granularity

Table 3 shows the cost spectrum for encoding a 7-function decision pipeline (loan approval) at three granularity levels, using a domain of 150 inputs (30 applicants $\times$ 5 loan amounts).

Table 3: Encoding granularity spectrum for a 7-function decision pipeline.

Granularity	Build (s)	Space (B)	Bits/elem	Exposed
Root ($p = 7$)	0.02	628	33.5	0
Intermediate ($p \approx 3$)	0.05	1,204	64.2	3
Leaf ($p = 1$)	0.08	2,116	112.9	7

The root encoding exposes zero intermediate values (maximum confidentiality) at 33.5 bits per element. The leaf encoding exposes all 7 intermediate cipher values (zero correlation hiding) at 112.9 bits per element—3.4$\times$ the root cost. The intermediate level balances the trade-off: 3 exposed values, 64.2 bits per element.

7.4 Case Study: Confidentiality Improvement

We demonstrate the three levers on a system with vocabulary $m = 10{,}000$ and Zipf-distributed queries.

The combined strategy improves e from 0.59 to 0.85: a 52% space overhead and 50% bandwidth overhead yields a 26 percentage point confidentiality improvement.

8 Discussion and Open Questions

The entropy ratio as a design tool. The entropy ratio e gives system designers a single number to optimize. Given a resource budget (space S, bandwidth B), the designer solves: maximize e subject to space $\leq S$ and bandwidth $\leq B$. The three levers (noise, multiplicity, granularity)

Table 4: Confidentiality improvement via three levers.

Configuration	e	Space overhead	Bandwidth overhead
Baseline (simple substitution)	0.59	1.0×	1.0×
+ Homophonic ($b = 100$)	0.77	1.52×	1.0×
+ Noise injection ($R/N = 0.5$)	0.72	1.0×	1.5×
Combined	0.85	1.52×	1.5×
Theoretical maximum	1.00	$\to \infty$	$\to \infty$

provide the degrees of freedom. Noise costs bandwidth but not space; multiplicity costs space but not bandwidth; granularity trades confidentiality for functionality.

Relationship to the orbit closure. The entropy ratio measures *passive* confidentiality: what the adversary learns by observing the cipher value stream. The orbit closure from [3] measures *active* confidentiality: what the adversary learns by applying the cipher maps it holds. A complete confidentiality analysis requires both: high e (passive) and small orbit (active). Connecting these two measures—showing that high e implies small effective orbit under certain conditions—is an open problem.

Adaptive $K(x)$. The multiplicity $K(x)$ is set at construction time based on an assumed distribution D. If D changes over time (e.g., query distribution shift), the system loses uniformity. Adaptive $K(x)$ that adjusts to observed frequencies without reconstruction would improve robustness. This requires an online construction strategy where representations can be added incrementally.

Tight bounds for composition. The general composition theorem gives $\eta_{g \circ f} \leq 1-(1-\eta_f)(1-\eta_g)$ under independence. For specific circuits (e.g., 3-SAT evaluation over cipher Booleans), the actual error rate depends on the input distribution and the circuit structure. Case-by-case interval arithmetic gives tighter bounds but does not scale. Automated analysis tools that propagate entropy through circuit DAGs would be useful.

Beyond Boolean search. This paper focuses on encrypted Boolean search as the primary application. The cipher map framework supports arbitrary functions $f : X \to Y$, and the confidentiality theory extends naturally. Ranked retrieval (scoring functions), phrase search (bigram models), and approximate string matching all admit cipher map implementations with different confidentiality profiles.

Limitations.

1. The entropy ratio is an average measure; it does not bound the leakage of any *specific* query. A single high-frequency query may be identifiable even when e is close to 1.

2. Compression-based estimation is positively biased (overestimates confidentiality). The bias decreases with sequence length but can be significant for short sequences.

3. The analysis assumes the random oracle model for cryptographic hashing. In practice, hash functions have biases that may reduce confidentiality below the theoretical predictions.

4. Marginal uniformity ($\delta \approx 0$) is necessary but not sufficient. Joint distributions, temporal patterns, and side channels (timing, bandwidth) can leak information that the entropy ratio does not capture.

9 Conclusion

We have developed a quantitative confidentiality theory for cipher map systems, grounded in the cipher map framework of [4]. The entropy ratio $e = H/H^*$ provides a principled, computable measure of how much the untrusted machine learns from the cipher value stream. Three levers—noise injection, multiple representations, and encoding granularity—offer distinct cost/benefit trade-offs for improving e, all expressible in terms of the cipher map parameters $(\eta, \varepsilon, \delta, \mu)$.

The key insight is that the cipher map's four properties already determine the confidentiality profile. Totality enables noise injection (filler queries are indistinguishable from real queries). Representation uniformity enables frequency hiding (multiple representations flatten the distribution). Composability enables chained evaluation but introduces correlation leakage. Correctness provides plausible deniability ($\eta > 0$ means the adversary cannot be certain of any decoded output).

Experimental validation on the 20 Newsgroups corpus confirms the theoretical predictions: FPR compounds as p_T^k for AND chains and $1-(1-p_T)^k$ for OR chains; the encoding granularity spectrum runs from 33.5 bits/element (root, zero intermediates) to 112.9 bits/element (leaf, all intermediates exposed); and the combined homophonic/noise strategy improves confidentiality from 59% to 85% with moderate overhead.

The theory connects two previously separate lines of work: the cipher map formalism (properties and composition) and the entropy-based confidentiality analysis (ratios and maximum entropy). Together, they provide a complete framework for designing, measuring, and improving confidentiality in systems that compute on data hidden behind a trapdoor.

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